Signals and Systems Operations

Participated in & did the coursework [Y/N]?

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1 INTRODUCTION

This laboratory exercise focuses on the fundamental operations of continuous time signals and systems. Specifically, the objective of the exercise is to mathematically define, simulate, and visualize operations of continuous time signals and systems such as cross-correlation and convolution using computational tools.

In order to attain the objective signals are represented mathematically using symbolic expressions. The Heaviside step function is useful in this context as it allows for the precise definition of piecewise continuous signals by acting as a mathematical switch [1]. To mathematically measure the similarity between these signals, cross-correlation is used. It measures the similarity between two distinct signals as one is shifted relative to the other. By evaluating the different cases of these overlapping intervals the features of a reference signal that aligns with the other signal over time is determined [2]. Meanwhile, convolution is used to determine how a system reacts to an input signal based on its impulse response. The resulting output response, y(t) = x(t) * h(t) provides a complete behavioral model of a linear time-invariant system [2].

Utilizing matlab to apply these concepts, several specific MATLAB functions are needed. First, the syms function which initializes symbolic variables. Second, heaviside() which constructs the discrete transitions in the signal amplitudes. Third, the int() function which computes the area under the overlapping signals. Fourth, subs() which replaces symbolic variables to evaluate specific time shifts. Moreover, visualization functions are needed to graphically interpret the results where functions such as fplot() to graph the symbolic expressions over specified domains, subplot() to display the different plots side-by-side, title() to add a descriptive title to the plots, and grid on to organize multiple graphs into clear, readable figures are used.

2 EXPERIMENTAL DESIGN

The experimental approach for this study centers on the computational modeling and verification of fundamental signal operations—specifically cross-correlation and convolution—using the MATLAB environment. By transitioning from symbolic mathematical definitions to graphical system responses, the design allows for an analytical comparison of how different signals interact over time. The methodology is structured into three primary phases: signal definition, operational segmentation, and computational execution.

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2.1 Mathematical Signal Definitions

The initial stage of the experiment involves the precise mathematical construction of the signals using the MATLAB Symbolic Math Toolbox. Signals are constructed using the `heaviside(t)` function to define specific temporal windows. For the cross-correlation phase, the "Target Signal" $x(t)$ is established as a ramp function bounded between $t = -2$ and $t = 2$, defined by $x(t) = t \cdot [u(t+2) - u(t-2)]$. The "Search Signal" $y(t)$ is a reflected and scaled ramp defined as $y(t) = -2t \cdot [u(t+1) - u(t-1)]$.

For the convolution phase, which simulates a system response, a pulse signal $x_{in}(t)$ is created using the expression $x_{in}(t) = u(t) - 2u(t-1) + u(t-2)$. This input is paired with a first-order system impulse response, $h(t) = e^{-10t}u(t)$, representing a rapidly decaying exponential characteristic.

2.2 Segmentation of Interaction Boundary Conditions

To calculate the total response, the experimenter must identify the critical time-shifts (τ) where the overlapping area between the signals changes its geometric behavior. The following table summarizes these segments to guide the symbolic integration process:

Case	Interaction Phase	Interval Boundary	Integration Limits
1	Pre-Interaction	$\tau < -3$	Integral = 0
2	Signal Entry	$-3 \le \tau < -1$	2 to $\tau + 1$
3	Full Immersion	$-1 \le \tau \le 1$	$\tau - 1$ to $\tau + 1$
4	Signal Exit	$1 < \tau \le 3$	$\tau - 1$ to 2
5	Post-Interaction	$\tau > 3$	Integral = 0

Table 1
Boundary conditions for cross-correlation integration.

2.3 Step-by-Step Computational Procedure

The following logic flow serves as a procedural roadmap for executing the experiment in a MATLAB environment:

1) **Initialize Variables:** Define the independent and shifting time variables using `syms t tau real`.

2) **Define Core Functions:** Code the mathematical models for $x(t)$, $y(t)$, and $h(t)$ using Heaviside functions.
3) **Calculate Cross-Correlation:** Implement the integral $R_{xy}(\tau) = \int x(t)y(t-\tau)dt$ using the `int` function for each of the five boundary cases.
4) **Synthesize Results:** Consolidate the individual case integrals into a single unified function using the `piecewise` command.
5) **Simulate System Output:** Execute the convolution integral $y_{out}(t) = \int x_{in}(\tau)h(t-\tau)d\tau$ for $t \geq 0$.
6) **Visualize Data:** Generate high-resolution plots using `fplot` with appropriate axis labels and titles.

2.4 Procedural Flowchart and Algorithmic Logic

The experimental design is visually summarized by the following logical progression:

Start $\rightarrow$ **Define Symbolic Signals** $\rightarrow$ **Boundary Identification** $\rightarrow$ **Symbolic Integration** $\rightarrow$ **Piecewise Reconstruction** $\rightarrow$ **Graphical Rendering** $\rightarrow$ **End**

The underlying algorithm for the convolution output response $y(t)$ is implemented as follows:

```
IF t < 0:
    Output = 0
ELSE IF t >= 0:
    Output = Integral of (x(tau) * exp(-10*(t-
        tau))) from 0 to t
Render Plot on Interval [-5, 5]
```

3 Analysis of Results

3.1 Modeling of Primary Symbolic Signals

The experimental procedure began with the mathematical definition of two distinct signals, $x(t)$ and $y(t)$. The first signal, $x(t)$, was constructed as a unit ramp function t bounded by the interval $[-2, 2]$, while the second signal, $y(t)$, was defined as a reflected and scaled ramp $-2t$ within the narrower interval $[-1, 1]$. By utilizing Heaviside step functions to define these temporal windows, the signals were established with infinite mathematical precision, avoiding the discretization errors inherent in numerical methods. The initial plots in Figure 1 confirmed the accuracy of these definitions, showing $x(t)$ as the stationary reference and $y(t)$ as the waveform to be shifted for the correlation process.

3.2 Evaluation of Cross-Correlation Boundary Cases

The cross-correlation $R_{xy}(\tau)$ was analyzed by evaluating the sliding integration of $x(t)$ and $y(t-\tau)$ across five specific boundary conditions. These cases represent the stages of signal interaction:

- **Case 1** ($\tau < -3$) and **Case 5** ($\tau \geq 3$): total separation with zero overlap.
- **Case 2** ($-3 \leq \tau < -1$) and **Case 4** ($1 \leq \tau < 3$): partial entry and exit of the signals.
- **Case 3** ($-1 \leq \tau < 1$): full overlap where $y(t)$ is entirely contained within the bounds of $x(t)$.

Following the identification of these cases, the integration of the product of the signals was performed for each interval. The resulting total cross-correlation function, rendered in Figure 2, combined these piecewise results into a single symmetrical waveform that effectively quantifies the degree of similarity between the two signals as a function of the time-lag τ.

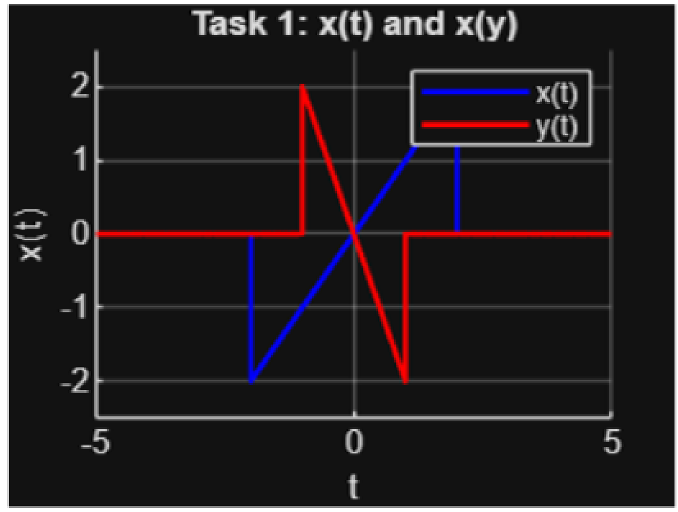

Figure 1. Primary Signal

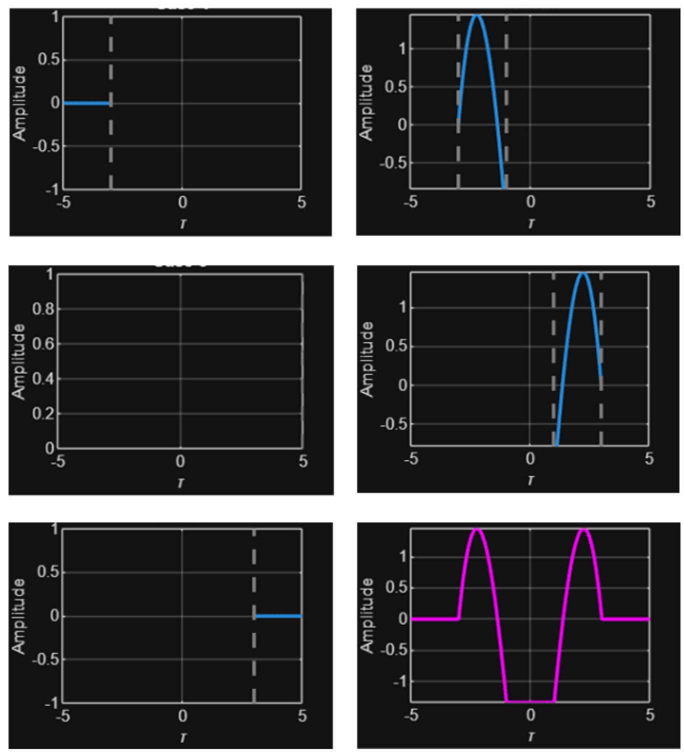

Figure 2. Cross-Correlation Cases and Total

3.3 LTI System Response via Convolution

The final stage of the analysis focused on the behavior of a Linear Time-Invariant (LTI) system subjected to a specific input pulse. The input signal $x_{in}(t)$ was defined as a pulse-like waveform $u(t)-2u(t-1)+u(t-2)$, and the system was characterized by an exponential impulse response $h(t) = e^{-10t}u(t)$. These signals were first plotted together in Figure 3 to visualize the system's initial state. The convolution was then evaluated through two temporal cases: **Case 1** ($t < 0$), where no interaction occurs, and **Case 2** ($t \geq 0$), where the input pulse is processed by the system's exponential decay. The final output response $y_{out}(t)$, as shown in Figure 3, exhibited a sharp rise followed by a rapid nonlinear decay, accurately reflecting the system's time constant and its causal response to the input stimulus.

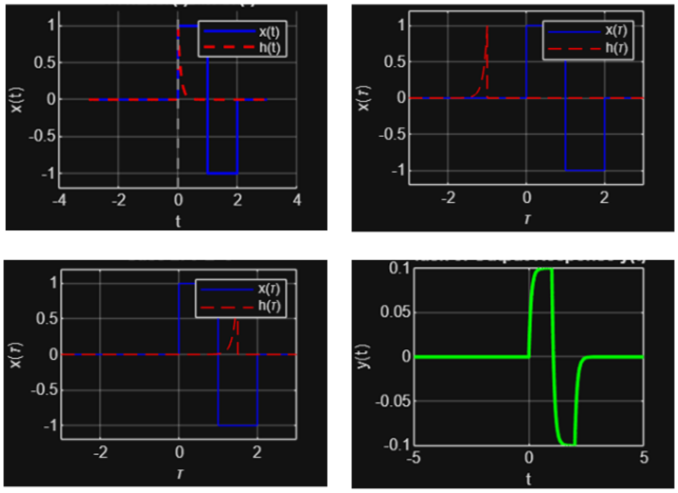

Figure 3. LTI System Cases of Convolution and Output

4 Discussion and Conclusion

The experimental results demonstrate the effectiveness of using symbolic computational tools to analyze fundamental signal operations. By using the MATLAB Symbolic Math Toolbox, the mathematical construction of signals $x(t)$ and $y(t)$ maintained infinite precision, successfully avoiding the discretization errors typically found in numerical methods. The analysis of cross-correlation $R_{xy}(\tau)$ specifically highlighted the critical role of boundary conditions in piecewise integration. As visualized in the transition from partial to full overlap in Cases 2 through 4, the resulting symmetrical waveform provides a precise quantification of signal similarity over the time-lag τ. This aligns with established theory that cross-correlation serves as a measure of the degree to which two signals are correlated as one is shifted relative to the other [1].

Furthermore, the study of Linear Time-Invariant (LTI) system responses through convolution revealed the distinct characteristics of causal system behavior. The interaction between the input pulse $x_{in}(t)$ and the exponential impulse response $h(t) = e^{-10t}u(t)$ resulted in an output $y_{out}(t)$ that accurately reflected the system's time constant. The sharp rise and subsequent rapid nonlinear decay confirm the system's immediate and fading memory response to an external stimulus [2]. Such behavior is a hallmark of first-order LTI systems, where the output is essentially a weighted integral of past inputs, governed by the specific decay rate of the impulse response [3].

In conclusion, this laboratory exercise successfully verified the theoretical underpinnings of signal interaction and system characterization. The structured methodology, moving from mathematical modeling to segmented computational execution, ensured that the resulting data was both mathematically grounded and highly accurate. The ability to visualize these complex interactions, such as the stages of signal passing and the processing of a pulse by an exponential decay, reinforces the importance of using computational logic to solve calculus-heavy signal processing tasks. These findings serve as a practical validation of the core principles of signals and systems, providing a reliable framework for analyzing more complex waveform interactions in future engineering applications.

5 Authors Contribution

On what contributions to specify, see the terms at www.elsevier.com.

1) BAUTISTA, Alyssa Nicole : Methodology, Software, Validation, Formal Analysis, Writing – Original Draft, and Review and Editing, Visualization, Project Administration.
2) GONZALES, Jeremia E.: Conceptualization, Software, Investigation, Resources, Data Curation, Writing – Original Draft, and Review and Editing, Supervision.
3) MENDIOLA, Keifer Judd: Methodology, Software, Validation, Formal Analysis, Investigation, Data Curation, Writing – Original Draft, and Review and Editing.

References

[1] A. V. Oppenheim, A. S. Willsky, and I. T. Young, *Signals and systems*, 2nd ed., ser. Prentice hall signal processing series. London: Prentice-Hall International, 1997.
[2] S. S. Haykin and B. Van Veen, *Signals and systems*. New York: Wiley, 1999.